

# **Information system for automatic detection of retinal diseases based on deep learning using optical coherence tomography images**

**8D06102 — Computer and Software Engineering**  
**Dissertation for the degree of Doctor of Philosophy (PhD)**

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- **Growing burden:** registered diabetes patients in Kazakhstan doubled in ten years — from 261 thousand (2014) to 517 thousand (2024); worldwide, 103 million people have diabetic retinopathy, 161 million by 2045.
- **Screening gap:** early retinopathy is asymptomatic, so every patient needs an annual fundus examination; the country has about 1,500 ophthalmologists —  $\approx 340$  diabetes patients per doctor; 36 % of the population is rural, but only  $\approx 10$  % of ophthalmologists work in rural areas.
- **Rationale:** the growing burden and the shortage of specialists call for automated screening from fundus images, but the accuracy of published models is unstable across cameras and acquisition conditions — a model is needed in which preprocessing is a component.

Sources: MoH RK (2024); Teo Z.L. et al., *Ophthalmology*, 2021; MoH RK (2026); Semenova Y. et al., *Hum Resour Health*, 2026; Bureau of National Statistics RK (2026).

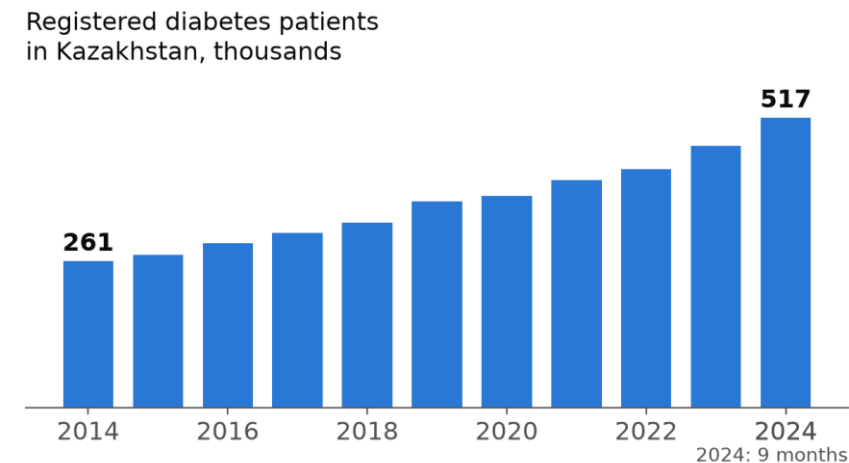


Figure 1 – Registered diabetes patients, 2014–2024

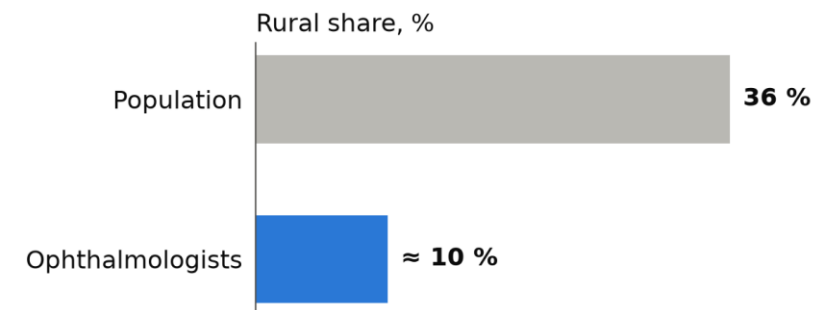


Figure 2 – Rural share: population 36 % / ophthalmologists  $\approx 10$  %

# Research aim and objectives

**Aim** — to develop and experimentally substantiate an integrated model for automated diagnosis of diabetic retinopathy from colour fundus images in which the preprocessing pipeline is a component of the model.

**Object of research** — the process of automated diagnosis of diabetic retinopathy from colour fundus images.

**Subject of research** — methods of fundus image preprocessing and of its integration with convolutional neural networks.

## Objectives

1. Analyse methods of automated diabetic retinopathy diagnosis and formulate the research problem
2. Develop the preprocessing pipeline and the integrated methodology: architectures, pretraining, interpretability, evaluation protocol
3. Evaluate the model on the source corpus and seven external corpora and validate the results statistically
4. Develop a screening system for settings without an accelerator and with intermittent connectivity

# Provisions submitted for defence

**Scientific novelty:** for the first time, fundus image preprocessing is formalised as a component of the diagnostic model,  $\hat{y} = \text{CNN}_{\theta}(\mathcal{P}(I))$ , and studied in a controlled factorial experiment: an eight-stage pipeline with a field-of-view mask as the fourth channel and a CLAHE clip limit defined as the minimum of two constraints is proposed, together with an asymmetric measure of agreement between model attention and expert annotation (ALO); the integrated model raises weighted F1 by 6.5 pp on both architectures studied.

## Provisions submitted for defence:

1. An eight-stage preprocessing pipeline for colour fundus images, formalised as a component of the diagnostic model and reducing differences between images in orientation, geometry, illumination and camera
2. An integrated "preprocessing pipeline + convolutional neural network" model with a four-channel input (RGB + field-of-view mask) for five-class diagnosis of diabetic retinopathy that improves accuracy on both architectures studied
3. An asymmetric measure of agreement between model attention and expert lesion annotation (ALO) for quantitative assessment of interpretability
4. A software system for diabetic retinopathy screening that reproduces the experimental pipeline and visualises the processing stages and attention maps

Table 1 – Analysis of existing approaches

| Direction                                                                                                                                                                                           | What is done                                                | Gap                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|
| Architectural (ensembles, CNN+ViT, attention)                                                                                                                                                       | accuracy grows through architectural complexity             | preprocessing is neither controlled nor reported     |
| Preprocessing (CLAHE, normalisation)                                                                                                                                                                | data preparation; default parameters                        | no ablation, parameters not formalised               |
| Cross-corpus transfer                                                                                                                                                                               | adaptation on target-domain data                            | the mechanism is not measured, only its consequence  |
| Interpretability (Grad-CAM)                                                                                                                                                                         | qualitative heat maps                                       | no quantitative measure of agreement with annotation |
| Deployed systems [58], [62]                                                                                                                                                                         | prospective trials (sensitivity 87.2 %, specificity 90.7 %) | preprocessing not disclosed                          |
| Common practice: $\mathcal{P}(I) \rightarrow \text{data} \rightarrow \hat{y} = \text{CNN}(\text{data})$ Proposed: $\hat{y} = \text{CNN}_{\theta}(\mathcal{P}(I))$ , $\mathcal{P}$ a model component |                                                             |                                                      |

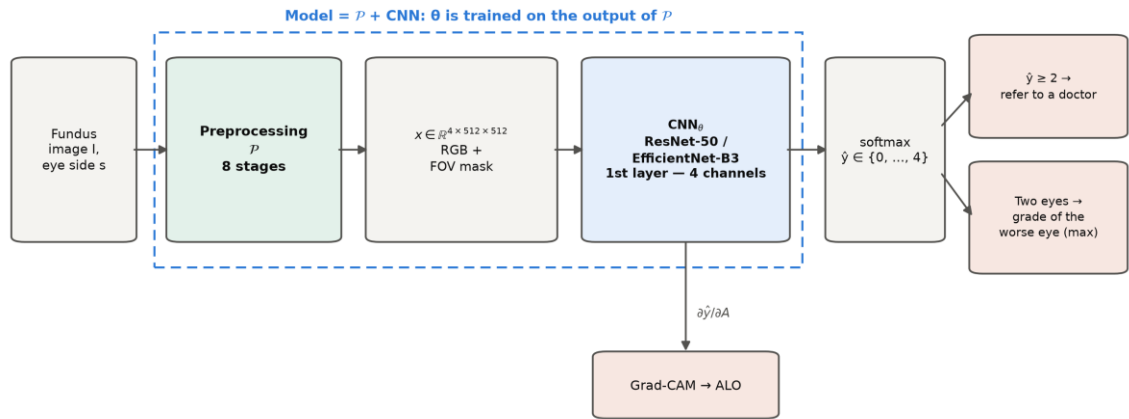


Figure 3 – The model as the composition  $\mathcal{P} + \text{CNN}$ , Grad-CAM  $\rightarrow$  ALO, two eyes  $\rightarrow$  grade of the worse eye

## Mathematical formulation

$$\mathcal{P} = T_7 \circ \dots \circ T_0 : (I, s) \mapsto x \in \mathbb{R}^{(4 \times 512 \times 512)} \tag{1}$$

$$\hat{y} = \operatorname{argmax}_k \operatorname{softmax}(\operatorname{CNN}_{\theta}(\mathcal{P}(I, s)))_k, k \in \{0, \dots, 4\}; \tag{2}$$

referral:  $\hat{y} \geq 2$

$$\theta^* = \operatorname{argmin}_{\theta} \sum_i \operatorname{FL}_{\gamma}(\operatorname{CNN}_{\theta}(\mathcal{P}(I_i, s_i)), y_i), \gamma = 2 \text{ — } \mathcal{P} \tag{3}$$

belongs to the model, not to the data

$$\text{ALO} = |A \cap L| / |L|, \text{ where } A \text{ is the Grad-CAM attention area} \tag{4}$$

and L the lesion mask

Table 2 – Mathematical formulation and architecture of the model

| # | Block                       | Content                                                                                                                                       |
|---|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Input                       | fundus image, left / right eye (s)                                                                                                            |
| 2 | Preprocessing $\mathcal{P}$ | 8 stages; deterministic at inference, stages 5–6 stochastic in training; output — $4 \times 512 \times 512$ tensor (RGB + field-of-view mask) |
| 3 | CNN backbone                | ResNet-50 (23.5 M) or EfficientNet-B3 (10.7 M), 4-channel first layer                                                                         |
| 4 | Classification              | softmax, $K = 5$ grades; referral: $\hat{y} \geq 2$                                                                                           |
| 5 | Patient level               | the patient's grade is the grade of the worse eye: $\hat{y} = \max(\hat{y}_{\text{left}}, \hat{y}_{\text{right}})$                            |
| 6 | Interpretation              | Grad-CAM (post hoc) $\rightarrow$ ALO, IoU                                                                                                    |

# Experimental design: $2 \times 2$ factorial

## Task 2

Table 3 – Experimental design:  $2 \times 2$  factorial

|                                                                                   | ResNet-50 | EfficientNet-B3 |
|-----------------------------------------------------------------------------------|-----------|-----------------|
| <b>Baseline:</b> stretch to $512 \times 512$ + ImageNet normalisation, 3 channels | <b>A</b>  | <b>C</b>        |
| <b>Integrated:</b> 8-stage pipeline, 4 channels                                   | <b>B</b>  | <b>D</b>        |

- Fixed training: Adam  $1 \cdot 10^{-4}$ , batch 16, 20 epochs (early stopping 5), focal loss  $\gamma = 2$ , 5 patient-level folds, seed 42
- Dominance criterion (fixed in advance):  $\Delta F1 \geq 5$  pp **and**  $\Delta AUC \geq 0.02$  **and** no drop in  $\kappa$
- Evaluation: DeLong, McNemar + Holm correction, mixed-effects ANOVA (arm  $\times$  architecture)

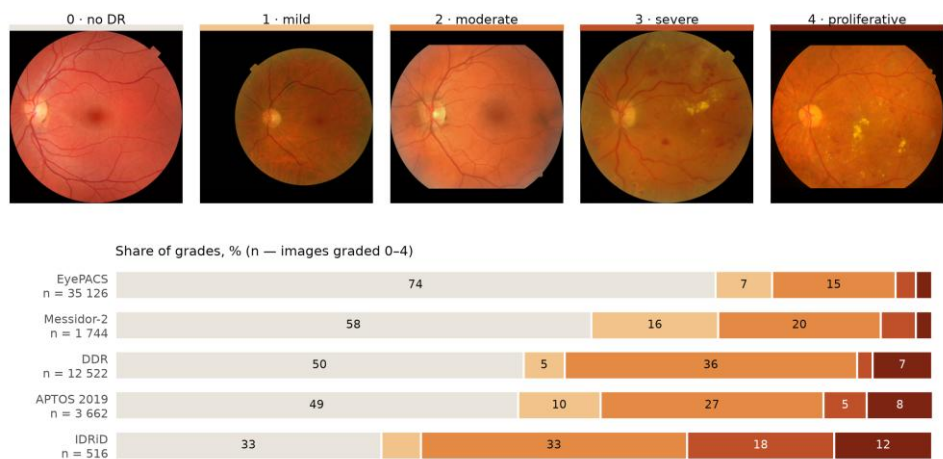
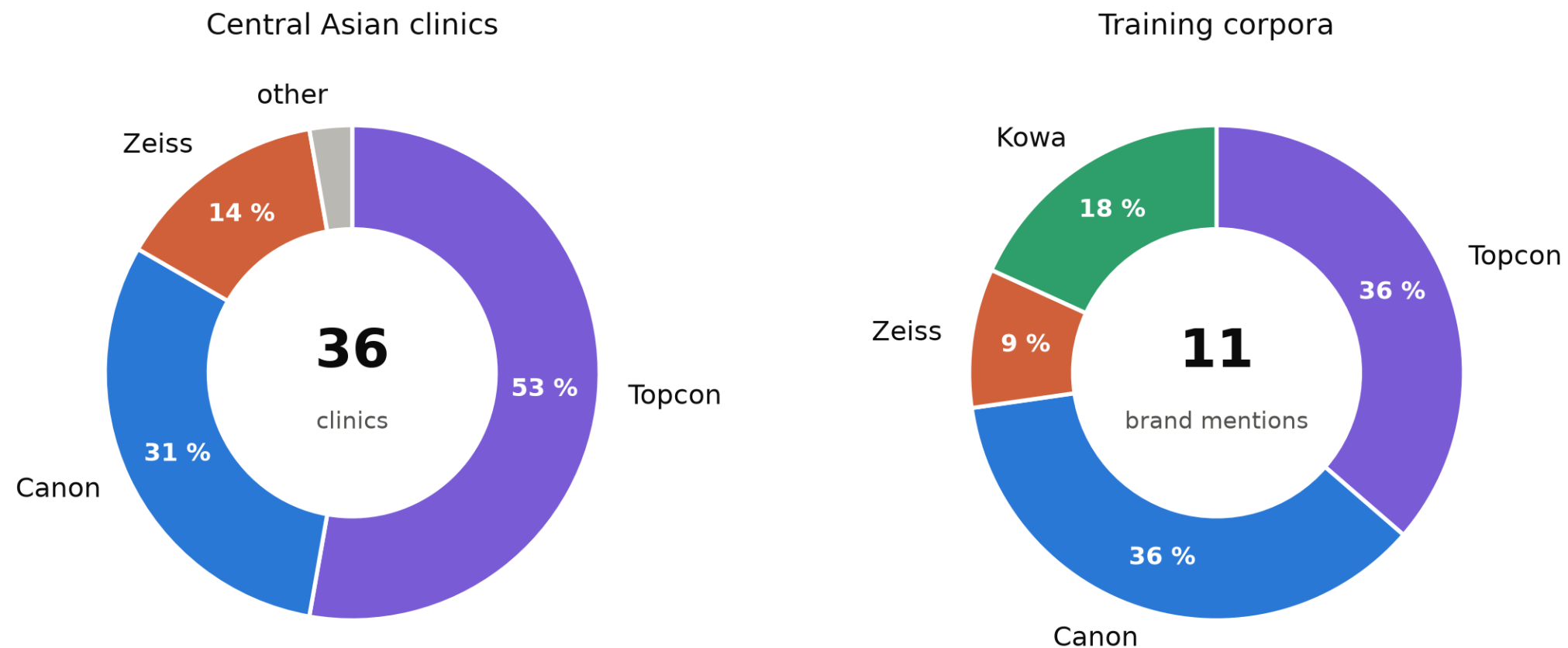


Figure 4 – Top — examples of DR grades 0–4, bottom — grade shares in the five-grade corpora

Table 4 – Datasets

| Corpus                | Group    | Role                              | Size                           | Camera                     |
|-----------------------|----------|-----------------------------------|--------------------------------|----------------------------|
| EyePACS               | training | tuning, in-domain evaluation      | ~35,126                        | Canon CR-1                 |
| APTOS 2019            | external | cross-corpus transfer             | ~3,662                         | mixed                      |
| Messidor-2            | external | external clinical performance     | ~1,748                         | Topcon TRC-NW6             |
| IDRiD                 | clinical | attention alignment, small sample | 516 (81 with masks)            | Kowa VX-10α                |
| Kazakhstan clinical   | clinical | held-out test                     | 60                             | —                          |
| DDR / ODIR-5K / RFMiD | device   | camera shift                      | ~13,673 / ~5,000 pat. / ~3,200 | Canon, Topcon, Zeiss, Kowa |

8 corpora · 4 camera manufacturers · DR 0 dominates EyePACS → focal loss, weighted F1



*Clinics: clinic websites and equipment suppliers; some rows give the most typical brand (central\_asia.md)*

Figure 5 – Two donut charts — camera brands in Central Asian clinics and in the training corpora, with a source note below

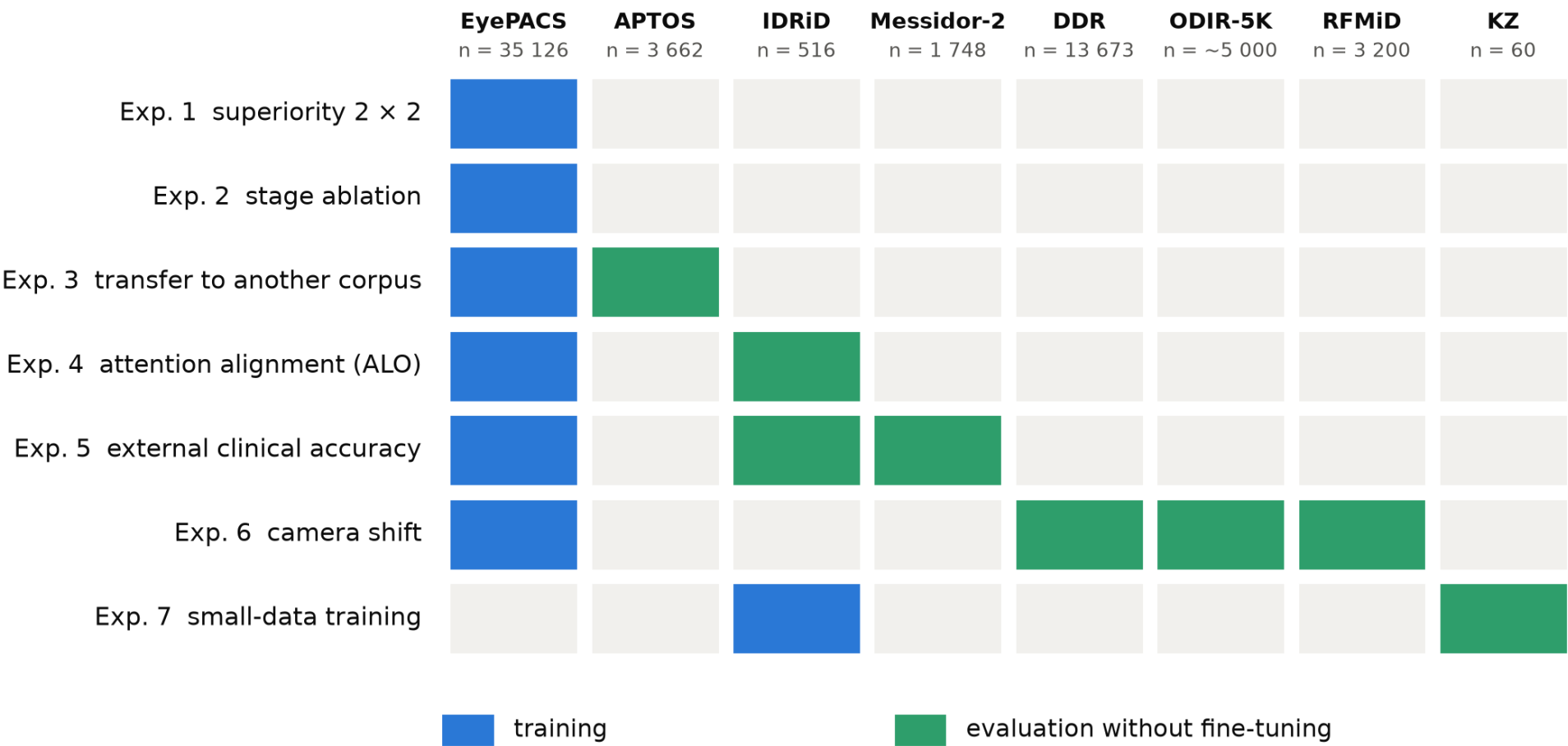


Figure 6 – Matrix of 7 experiments × 8 corpora; blue — training, green — evaluation without fine-tuning

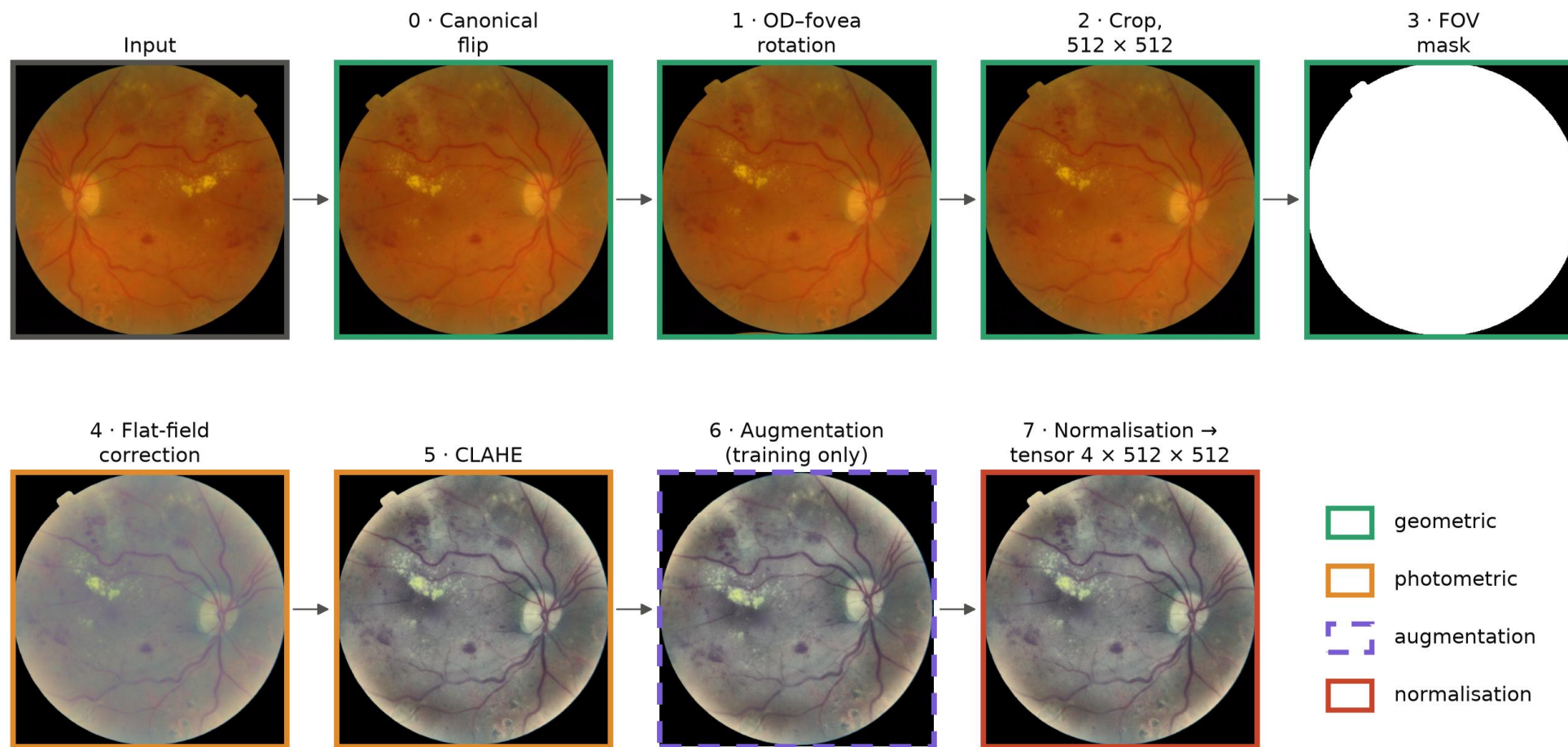


Figure 7 – One eye (DR03, left) through all 8 stages — the pipeline strip, groups colour-coded

Left eye — input:  
optic disc on the left



After the flip:  
disc on the right, as in a right eye



Figure 8 – The left eye — input (disc on the left) and after the flip (disc on the right)

# Rotation by optic disc and fovea (midpoint)

## Task 2

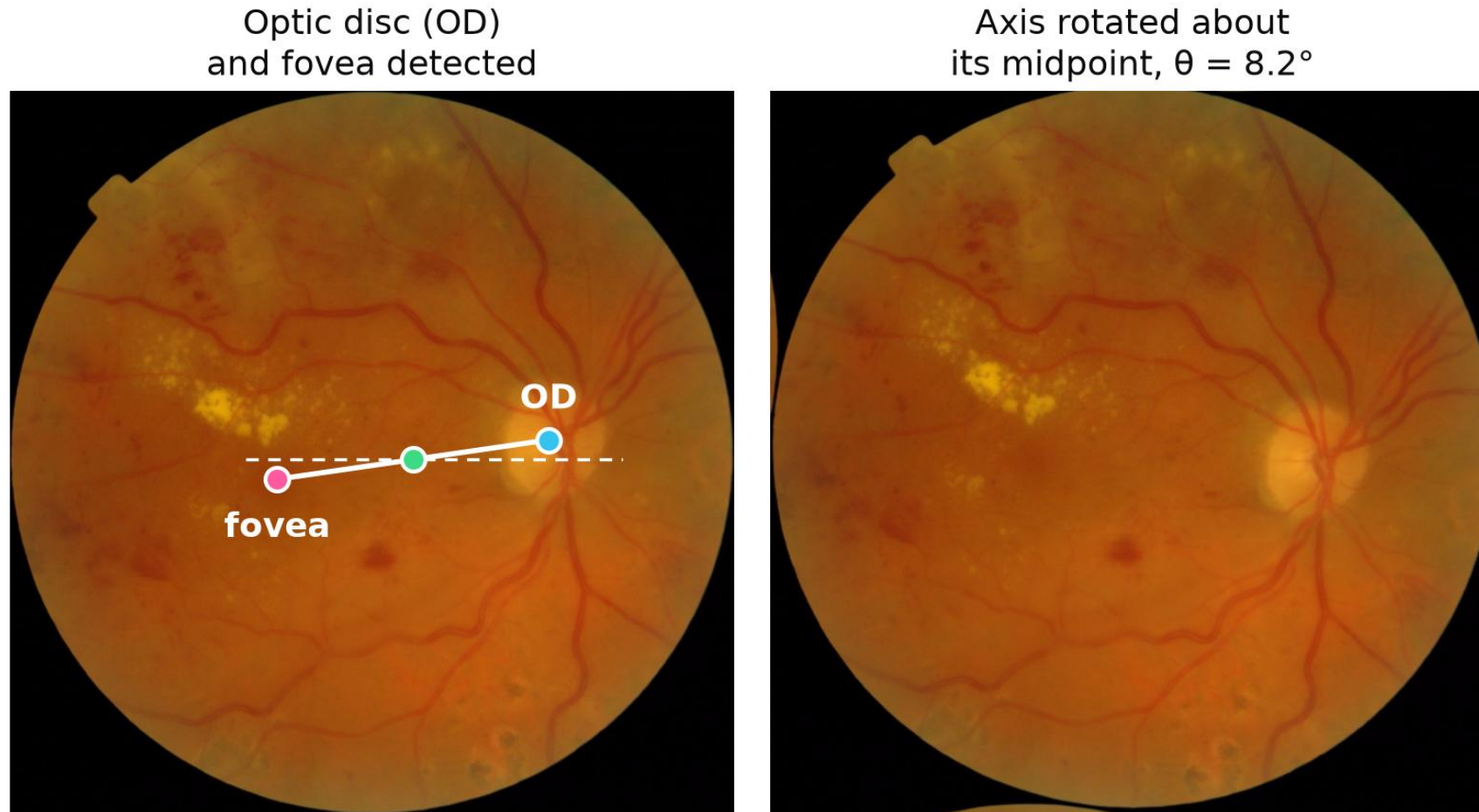
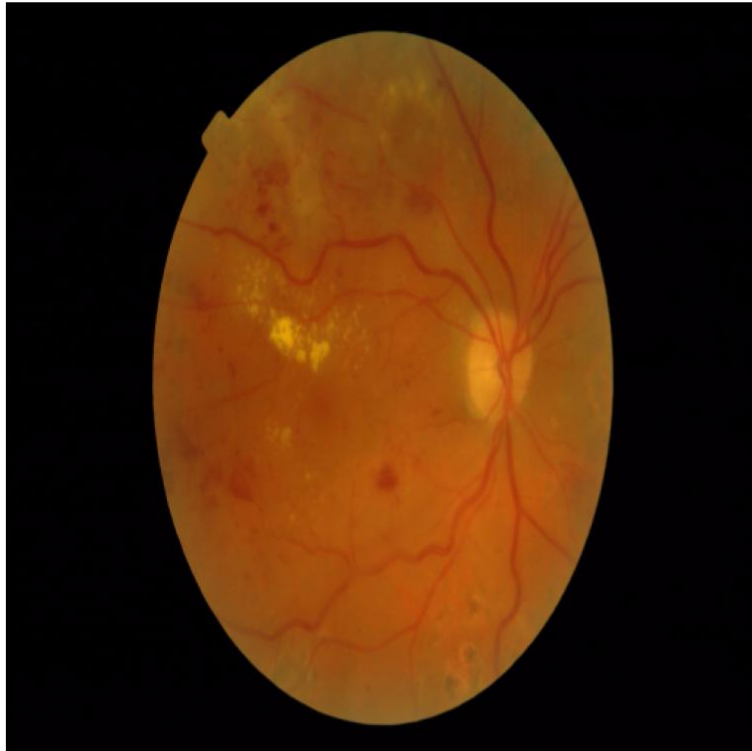


Figure 9 – The detected disc and fovea and the image rotated about the midpoint of the axis,  $\theta = 8.2^\circ$

# Crop & resize, field-of-view (FOV) mask

## Task 2

Baseline: stretch to  $512 \times 512$   
(the fundus becomes an ellipse)



Stage 2: crop and  
isotropic scaling



Stage 3: FOV mask  
(4th channel)

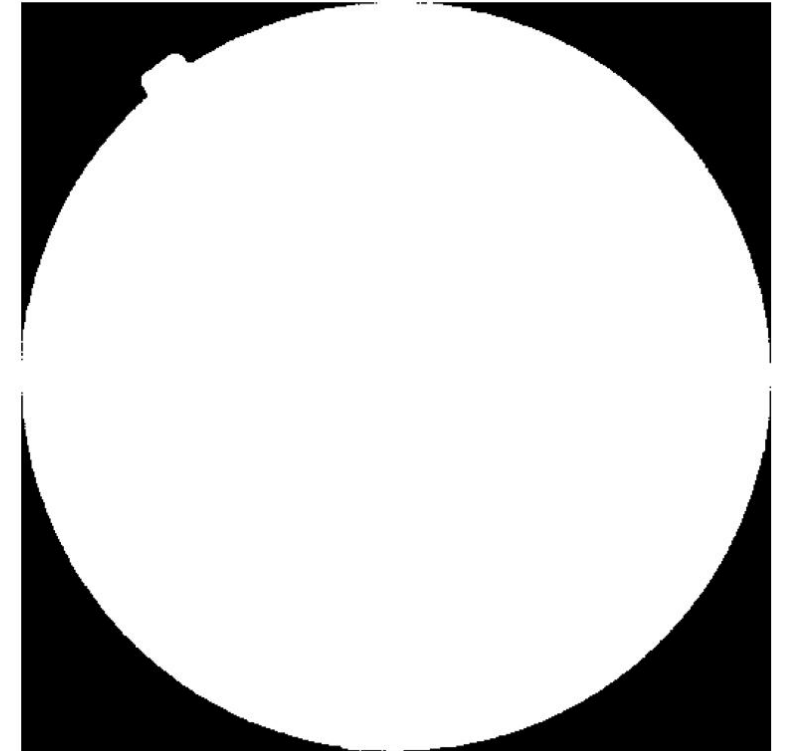


Figure 10 – The baseline stretch (the fundus becomes an ellipse) · stage 2 — isotropic crop · stage 3 — FOV mask

After cropping

4 · Flat-field correction  
 $\sigma = 0.07 \cdot D$

5 · Polar CLAHE  
(LAB, L channel)

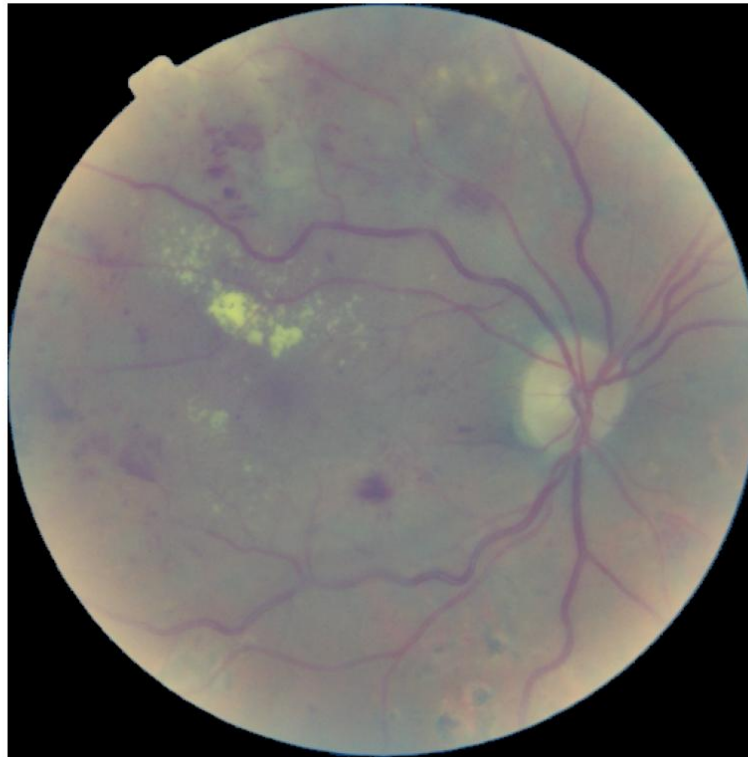


Figure 11 – After cropping → stage 4, flat-field correction ( $\sigma = 0.07 \cdot D$ ) → stage 5, polar CLAHE

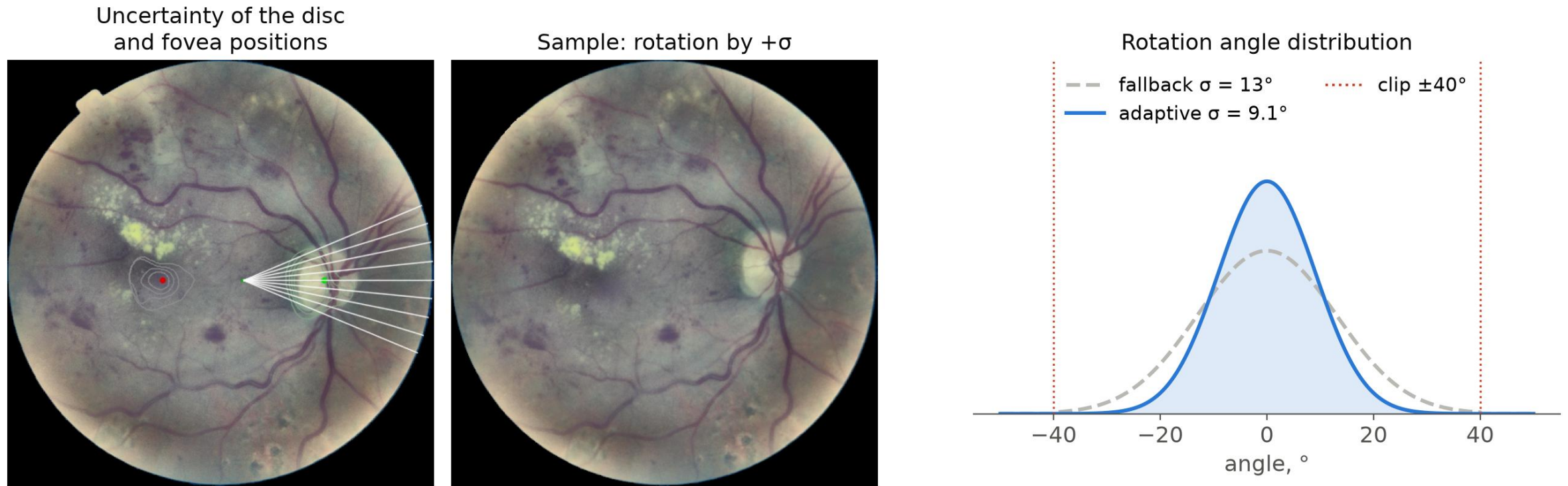
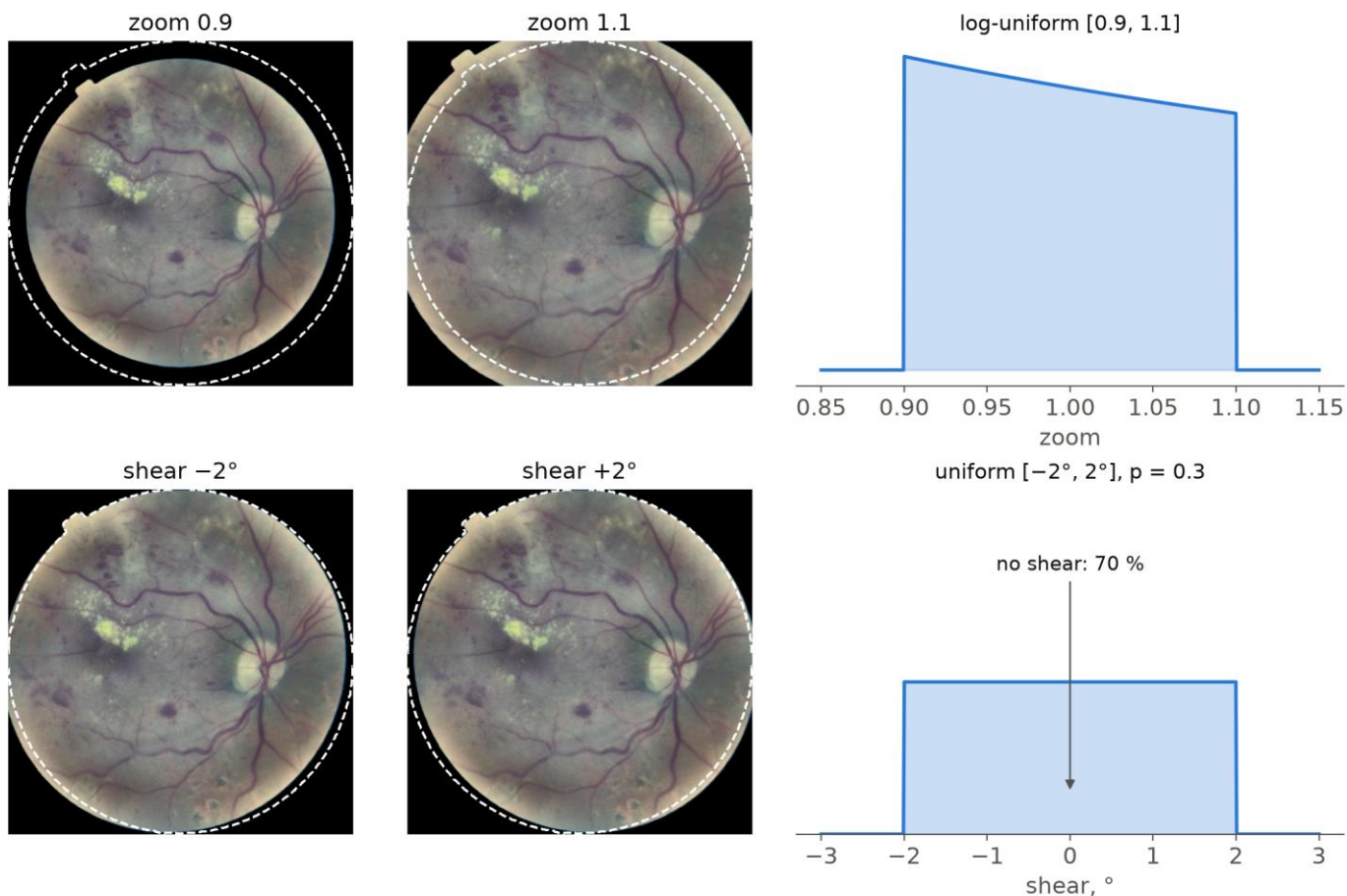


Figure 12 – Uncertainty of the disc and fovea positions · a sample rotated by  $+\sigma$  · angle distribution: adaptive  $\sigma = 9.1^\circ$  vs fallback  $13^\circ$ , clip  $\pm 40^\circ$



## Parameters:

- zoom — log-uniform  $[0.9, 1.1]$ , stretch  $[1/1.05, 1.05]$
- shear — uniform  $[-2^\circ, 2^\circ]$ ,  $p = 0.3$

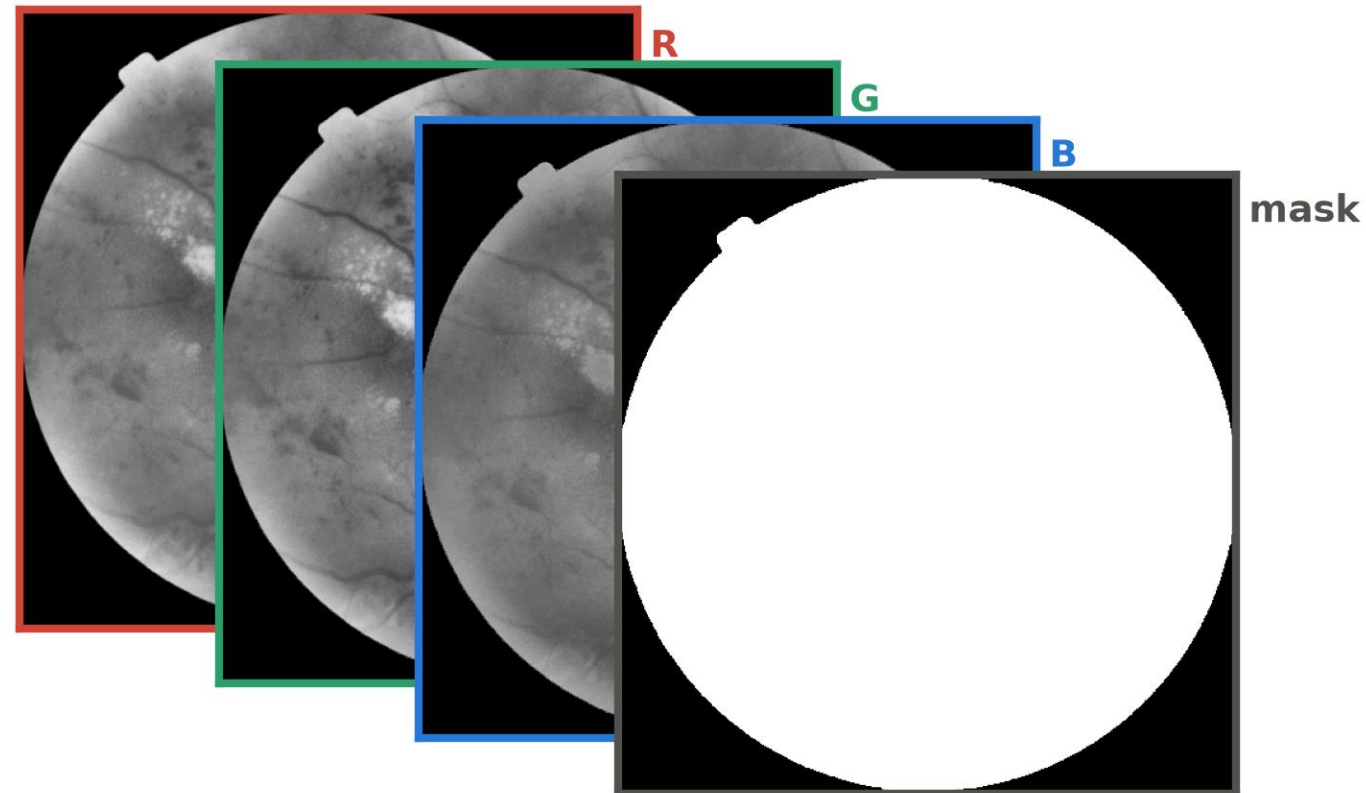
Figure 13 – Zoom 0.9 / 1.1 and shear  $\pm 2^\circ$  — extreme samples (dashed — the original field of view) and distributions



Figure 14 – Original · colour jitter min / max · noise + JPEG (4 $\times$  zoom)

## Parameters:

- brightness, contrast, saturation  $\times [0.9, 1.1]$ , hue  $\pm 0.02$  (each  $p = 0.5$ )
- noise  $\sigma = 2-6$  ( $p = 0.15$ )
- JPEG  $q = 70-100$  ( $p = 0.2$ )



$$x'_c = (x_c - \mu_c) / \sigma_c, \mu, \sigma \text{ from the valid pixels of EyePACS} \\ \text{tensor } 4 \times 512 \times 512$$

Figure 15 – The network input — a  $4 \times 512 \times 512$  tensor: normalised R, G, B and the mask;  $x'_c = (x_c - \mu_c) / \sigma_c$

# Training parameters: focal loss, optimiser, 5-fold CV; metrics: F1, AUC, Cohen's $\kappa$ , G, ALO

## Task 3

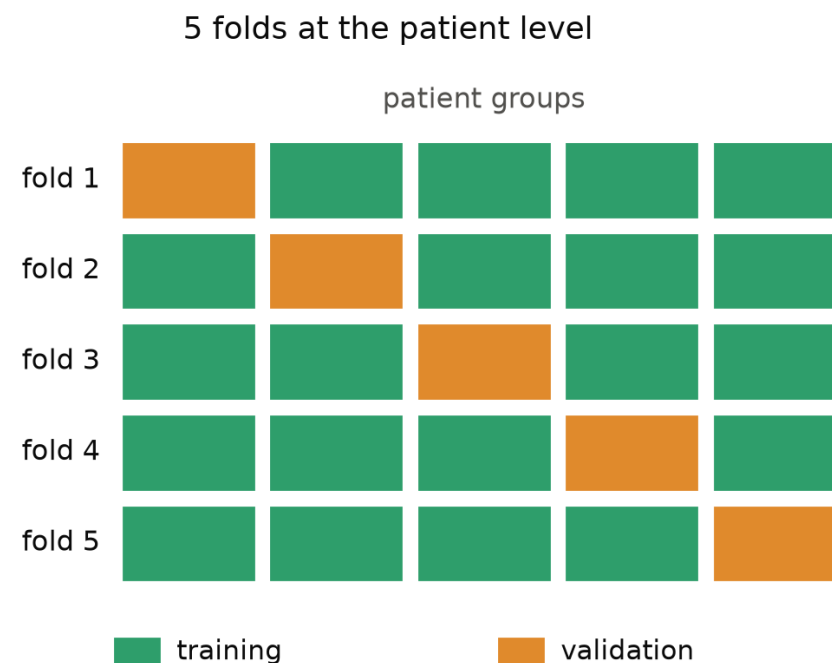
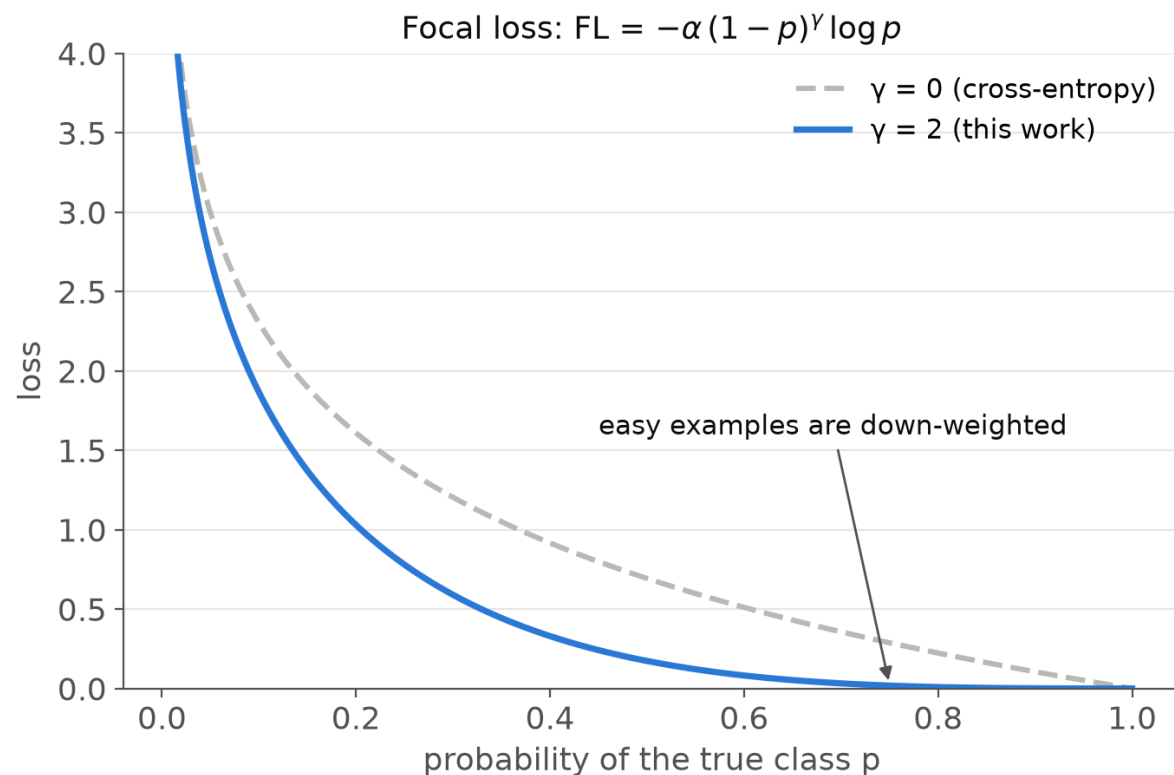


Figure 16 – Focal loss ( $\gamma = 2$  vs cross-entropy) and 5 patient-level folds

- Metrics: weighted F1 (primary), AUC, quadratic  $\kappa$
- On external corpora:  $G = F1_{\text{ext}} / F1_{\text{EyePACS}}$  (threshold 0.85); attention:  $ALO = |A \cap L| / |L|$

# Experiment 1: contribution of preprocessing across two architectures

Task 3

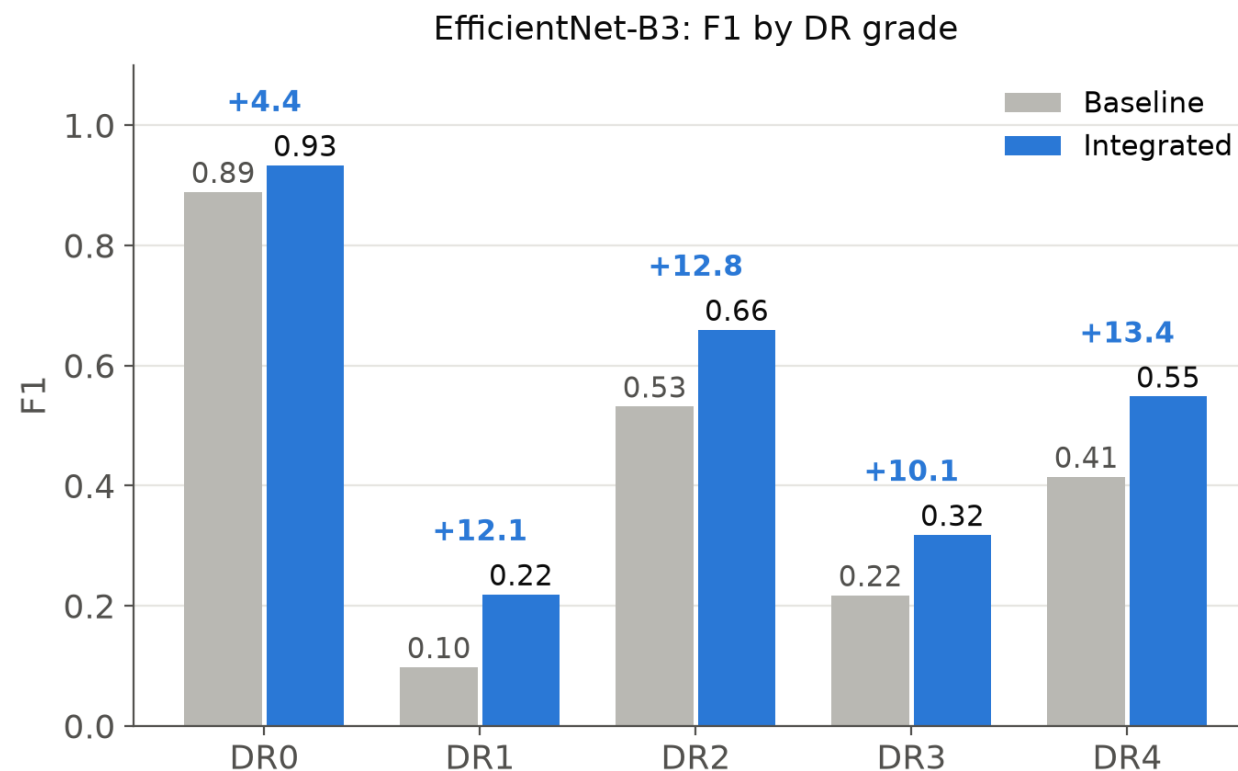
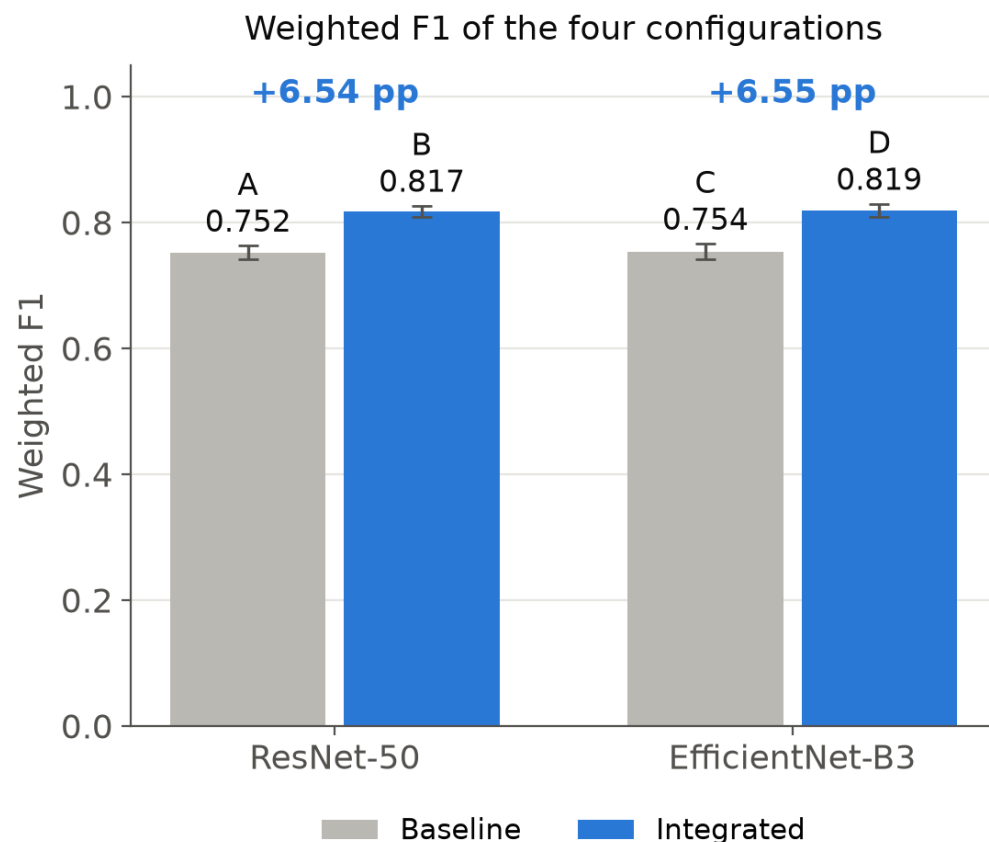


Figure 17 – Weighted F1 of the four configurations (A–D, SD over 5 folds); right — EfficientNet-B3, F1 by DR grade

# Experiment 2: ablation of pipeline stages

## Task 3

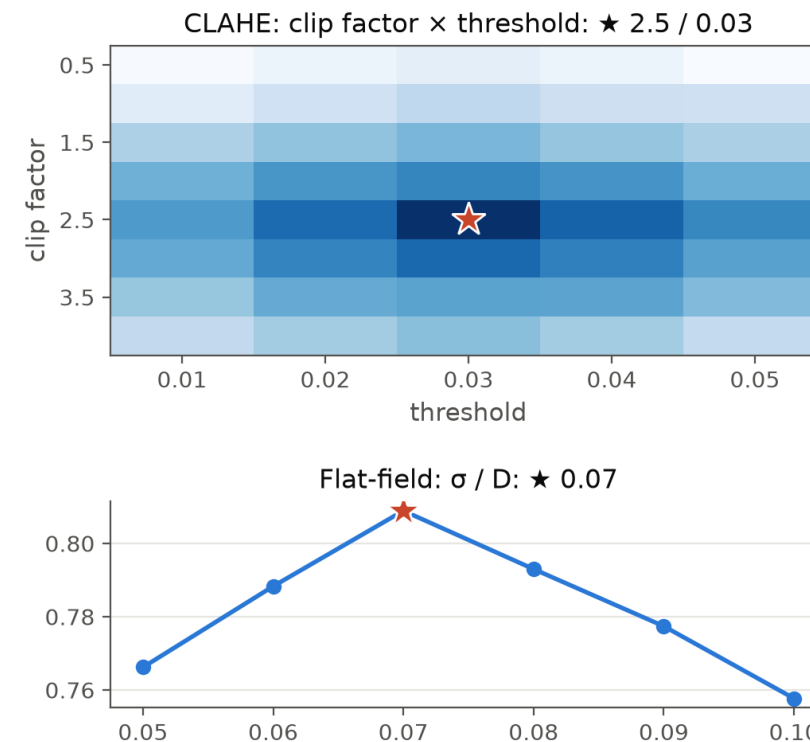
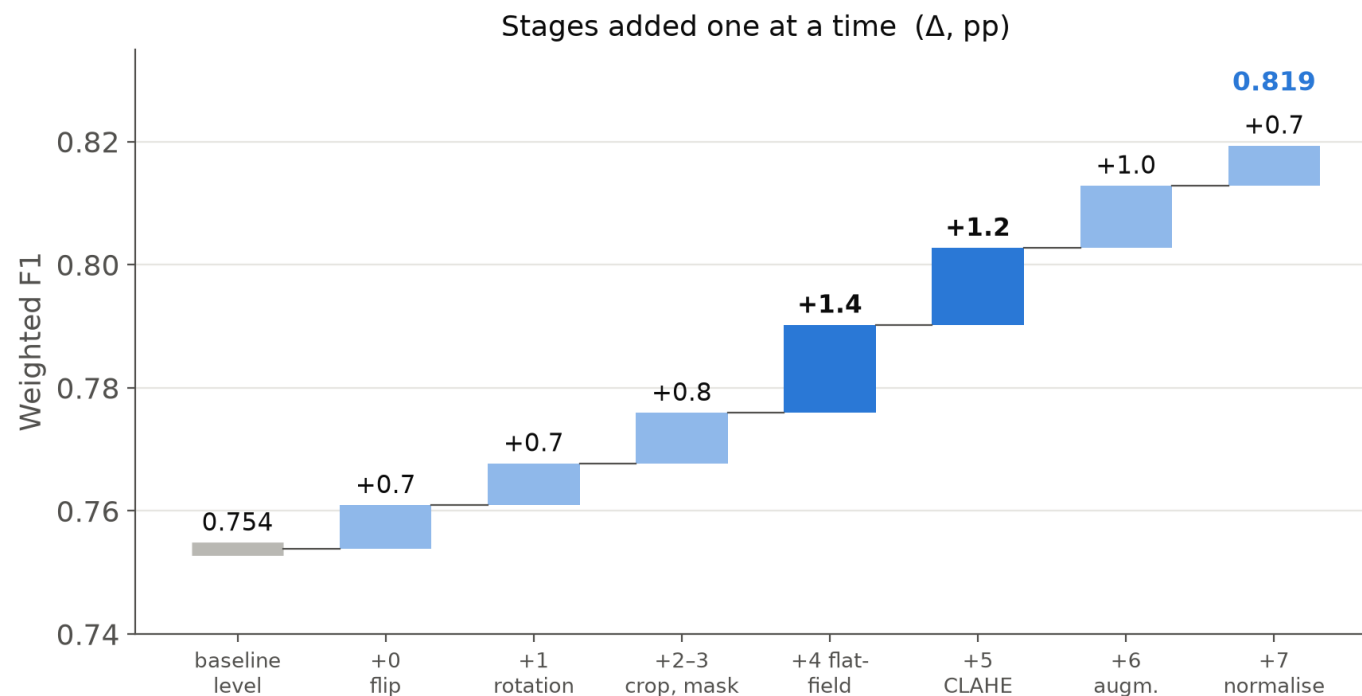


Figure 18 – Stages added one at a time (waterfall,  $\Delta$  pp); right — optimum of CLAHE (clip factor 2.5, threshold 0.03) and of flat-field ( $\sigma/D = 0.07$ )

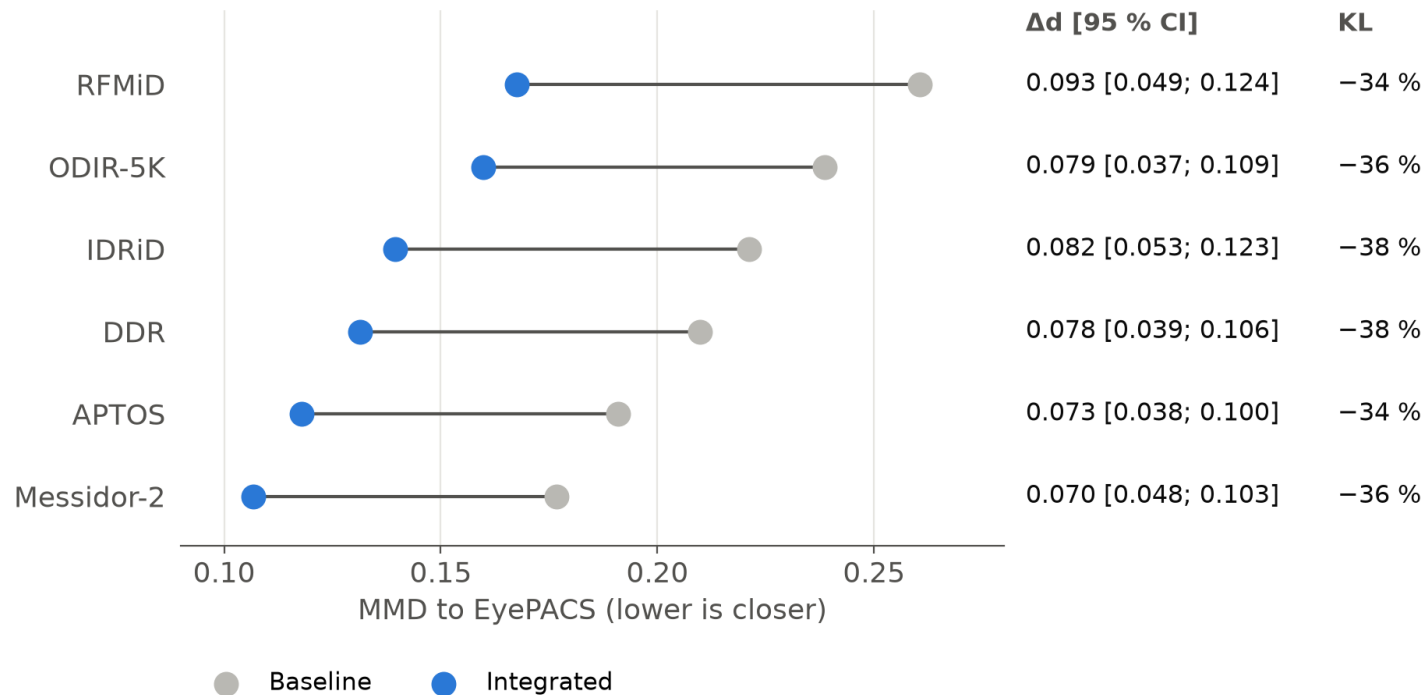


Figure 19 – MMD to six corpora — baseline and integrated; at the right  $\Delta d$  [95 % CI] and the KL reduction

- Distance is measured at the penultimate layer (MMD) and at pixel level (KL)
- It falls on **6 / 6** corpora:  $\Delta d = \mathbf{0.070...0.093}$ , every 95 % interval excludes zero; KL **-34...-38 %**
- Normalisation uses source-domain statistics only → the convergence is a property of preprocessing, not of adaptation

# Experiment 3: transfer to another dataset

## Task 3

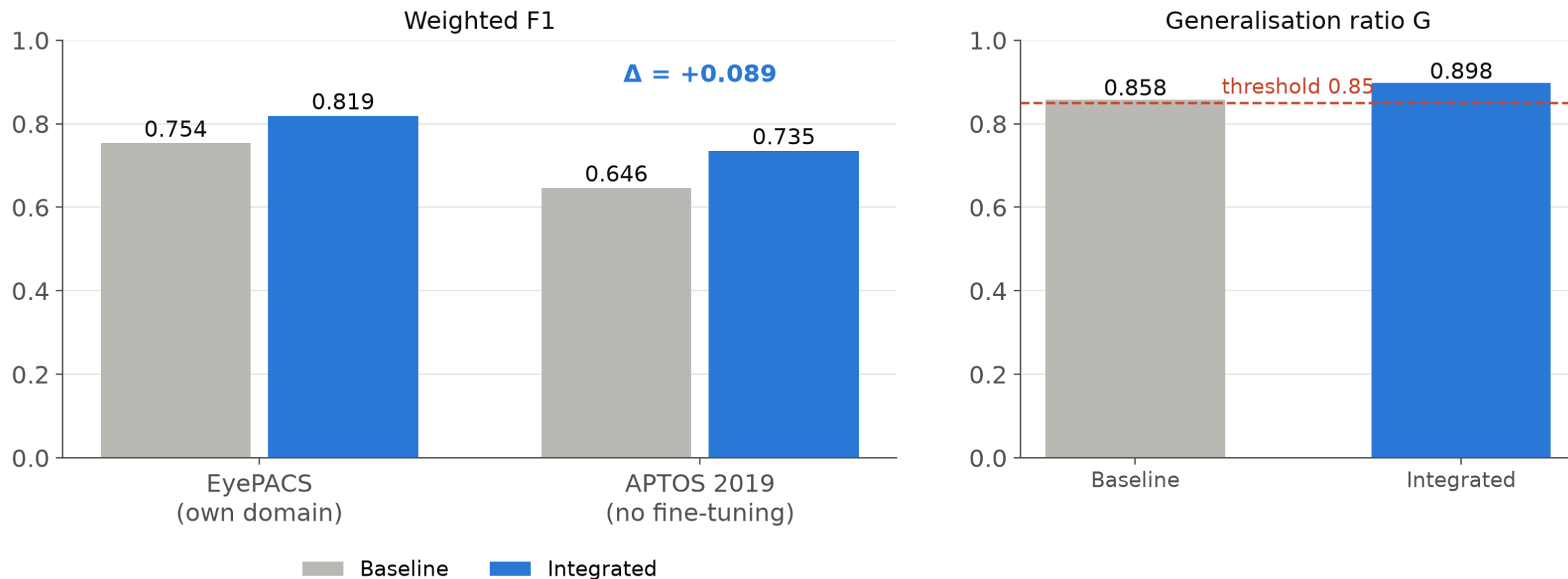


Figure 20 – Weighted F1 on EyePACS and APTOS 2019; generalisation ratio G, threshold 0.85

# Experiment 4: interpretability (ALO, Grad-CAM)

## Task 3

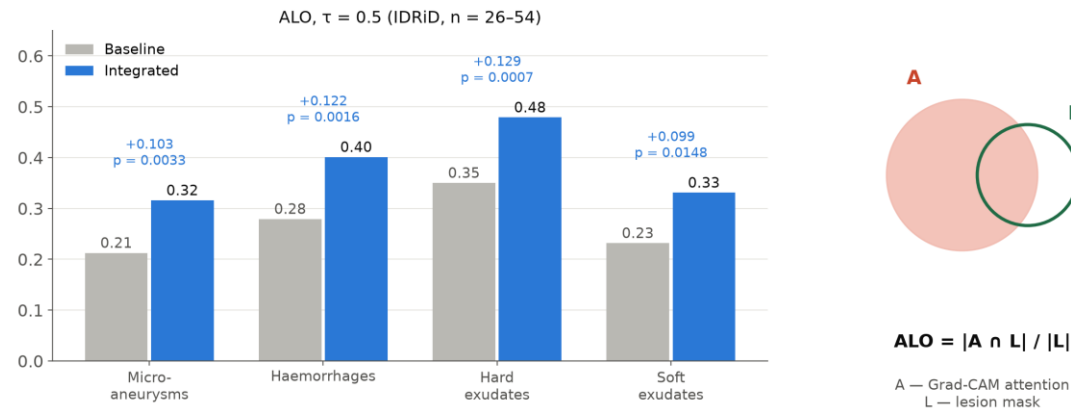


Figure 21 – ALO by lesion type and the ALO definition

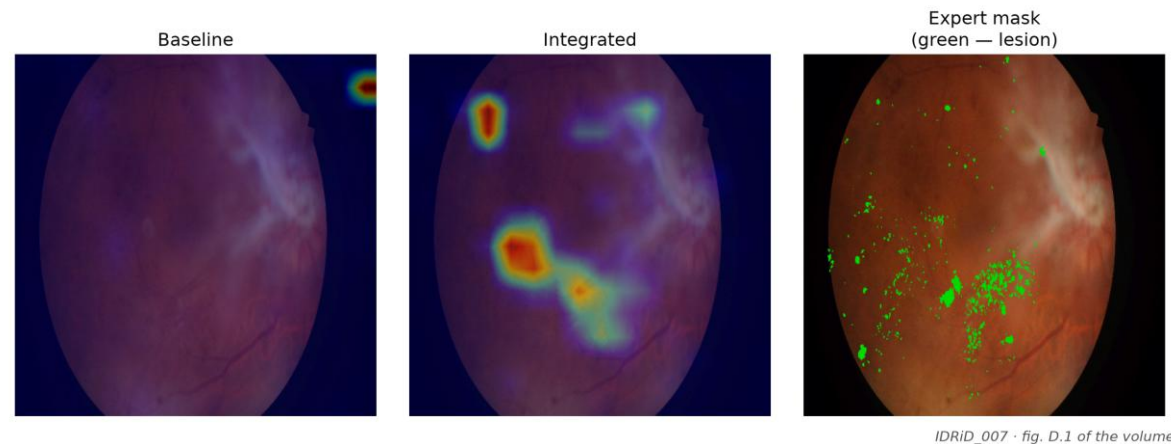


Figure 22 – The Grad-CAM pair: baseline · integrated · expert mask, IDRiD\_007 — fig. D.1 of the volume

# Experiment 5: accuracy on external clinical data

## Task 3

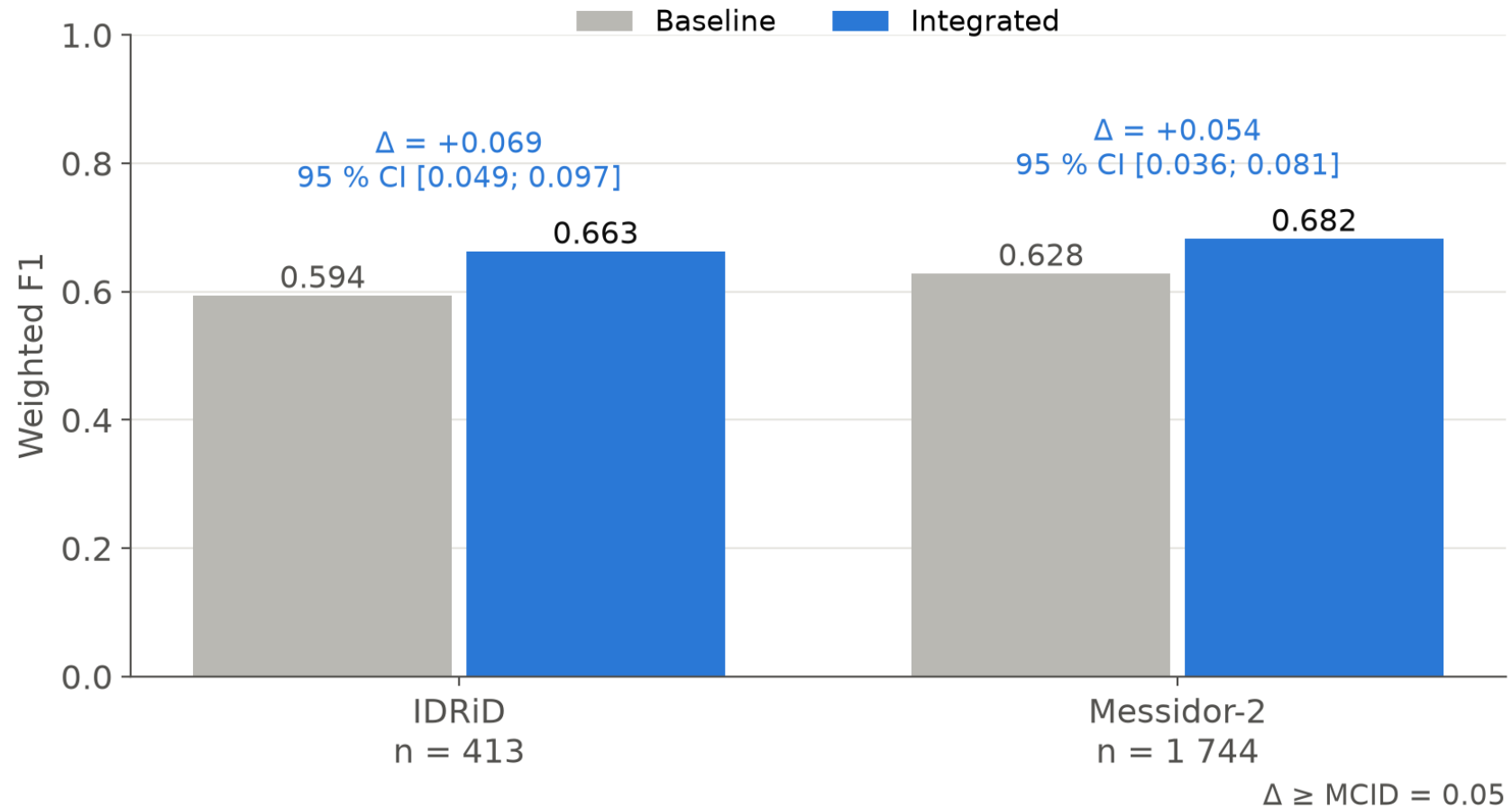


Figure 23 – Weighted F1 on IDRI and Messidor-2,  $\Delta$  and 95 % CI; MCID = 0.05

# Experiment 6: robustness to camera change

## Task 3

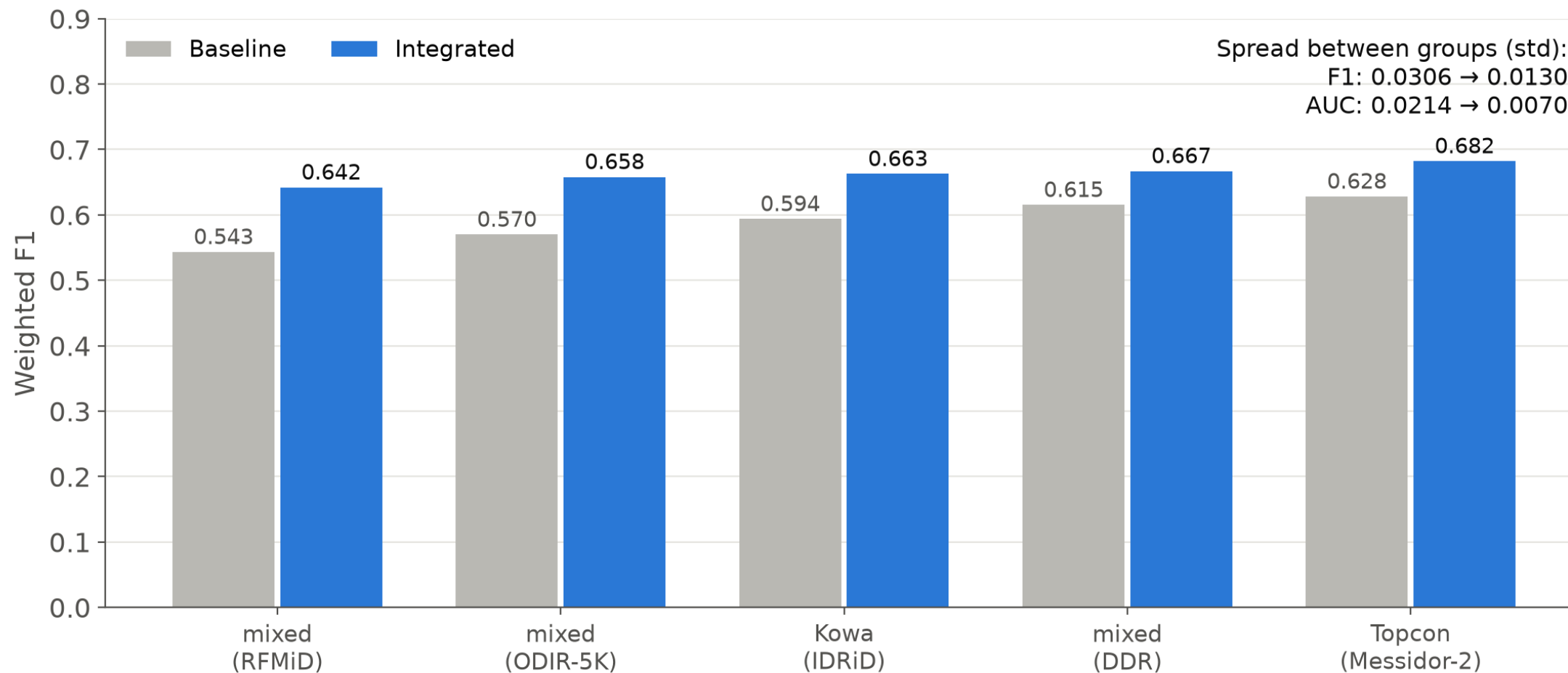


Figure 24 – Weighted F1 by the five camera groups and the spread between groups

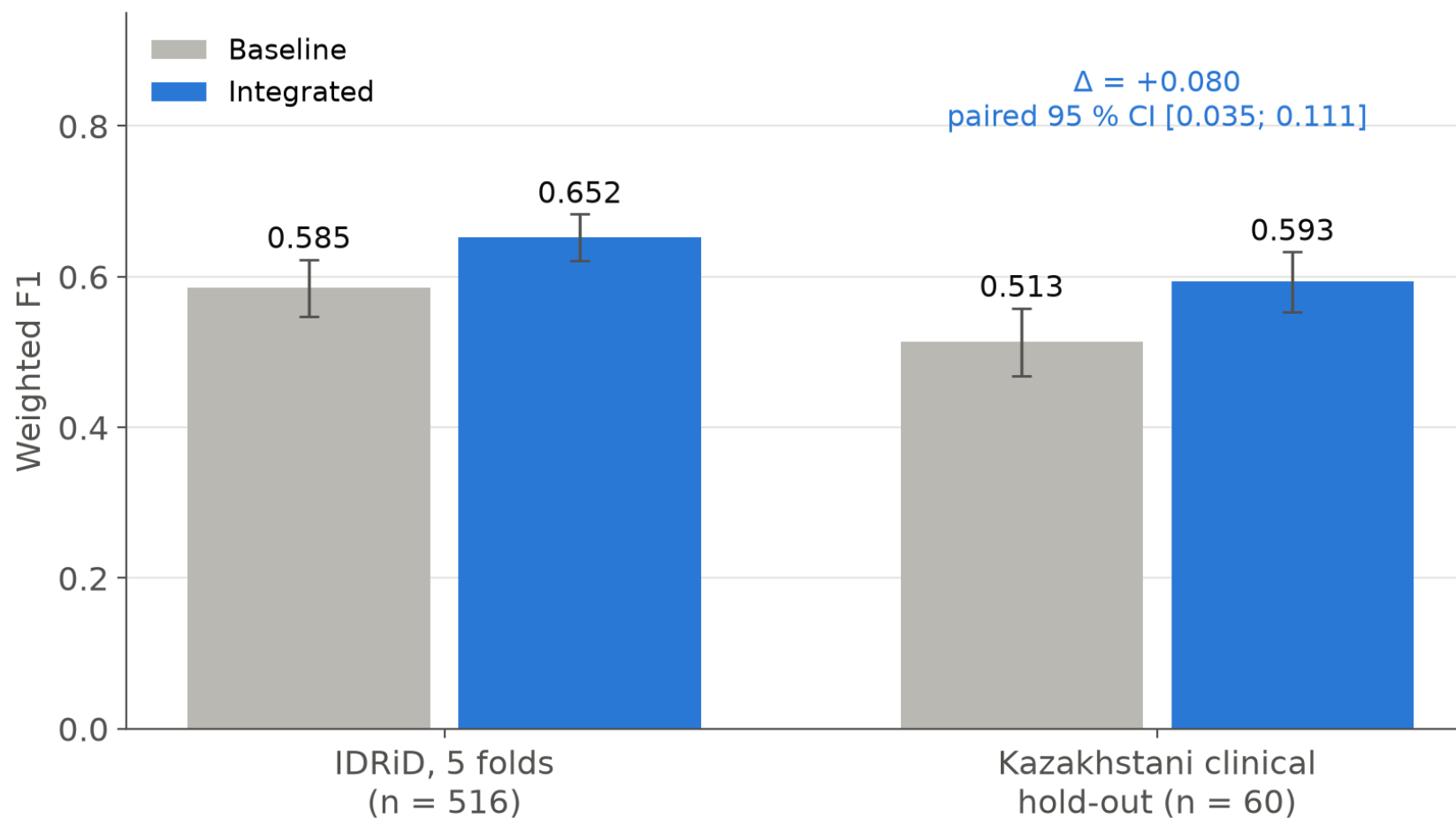


Figure 25 – IDRiD, 5 folds and the Kazakhstani clinical hold-out (n = 60): weighted F1, paired  $\Delta$  and 95 % CI

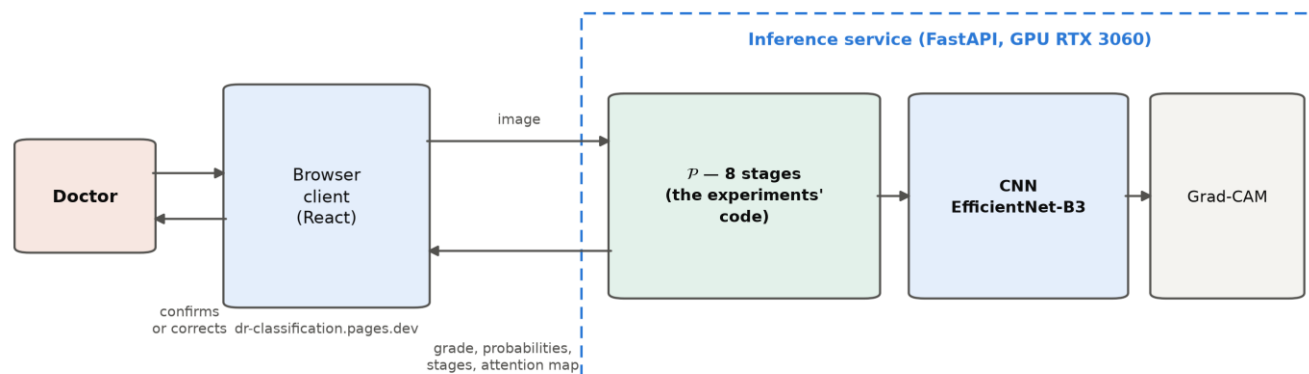


Figure 26 – System diagram: doctor ↔ browser client ↔ inference service ( $\mathcal{P} \rightarrow \text{CNN} \rightarrow \text{Grad-CAM}$ )

1. **Inference service + browser client**, 8 modules; the same pipeline code as in the experiments
2. **Shows every step**: preprocessing stages, grade, Grad-CAM attention map; the clinician confirms or corrects the decision
3. **On consumer hardware**: RTX 3060 12 GB; the 4th channel adds only +0.9 % FLOPs and +24 MiB memory
4. **Demonstrator**: dr-classification.pages.dev

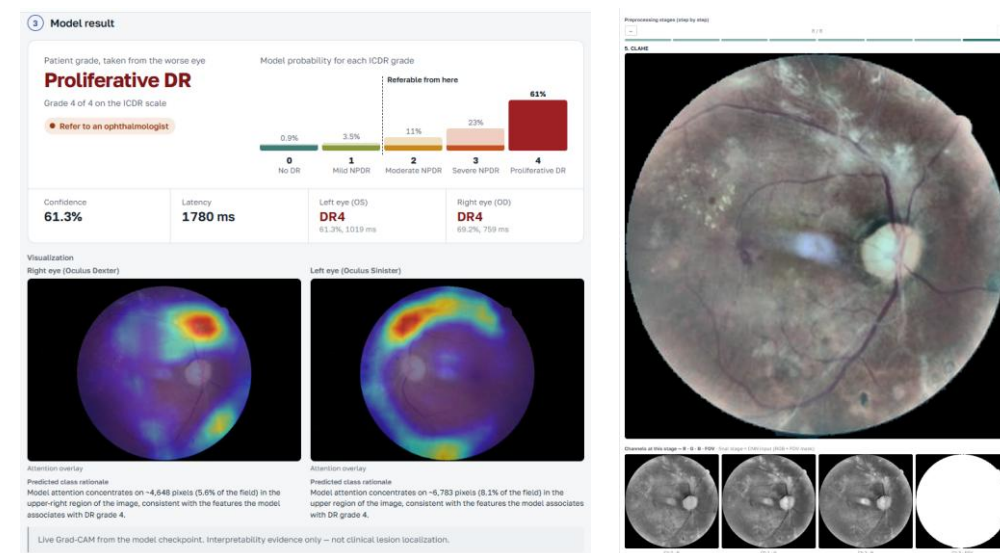


Figure 27 – Two screenshots of the browser client: (a) grade, class probabilities and Grad-CAM; (b) preprocessing stages

# Conclusion

- 1. Analysis (Ch. 1):** the research problem was formulated — preprocessing is not reported as part of the model; the requirements of resource-limited screening were defined
- 2. Methodology (Ch. 2):** the 8-stage pipeline was formalised (clip limit, illumination correction), two architectures with 4-channel input, pretraining selected by a gate ( $\kappa$  0.655 / 0.681), the evaluation protocol fixed before the experiments
- 3. Experiments (Ch. 3):** F1 **+6.5 pp**; ablation 0.754  $\rightarrow$  0.819; distance shrinks on 6 / 6 corpora; APTOS **+0.089**; camera spread **2.4×** narrower; IDRiD / Messidor-2 **+0.069** / **+0.054**; ALO 4 / 4; small data **+0.080** — all 7 pre-registered hypotheses supported
- 4. System (Ch. 4):** the inference service and browser client were built on the same pipeline code as the experiments; certificate No. 78109

**The main finding is consistency:** the advantage exists in-domain, decomposes across stages, goes with a measured distance and appears on every external corpus; the differences are confirmed by DeLong, McNemar and bootstrap

Total publications: 5

Scopus journal (Q3, CiteScore 2.5, 42nd percentile) — 1:

- Sapakova S., **Yesmukhamedov N.**, Sapakov A. Development of an Image Quality Enhancement Approach for Diabetic Retinopathy Diagnosis // Eastern-European J. of Enterprise Technologies. — 2025. — Vol. 4, No. 9(136). — P. 79–88.

CQAES (KOKSNVO) journals — 3:

- Yesmukhamedov N.S.** et al. Methods for Preprocessing and Analysis of Fundus Images... // Herald of KBTU. — 2025. — No. 4(75). — P. 119–130.
- Yesmukhamedov N.S.** et al. Development of an IS Architecture for Healthcare... // News of NAN RK. — 2025. — No. 2(354). — P. 74–91.
- Sapakova S.Z., Daniyarova D.R., **Yesmukhamedov N.S.** et al. Mathematical Modeling of Laser Exposure... // Вестник КазУТБ. — 2025. — Vol. 2, No. 27-740. — P. 20–30.

Scopus conference — 1:

- Sapakova S., **Yesmukhamedov N.** et al. Methods for Pre-processing and Analysis... // Procedia Computer Science (Elsevier). — 2025. — Vol. 272. — P. 496–501. (DS-2025, Istanbul)





Figure 28 – Certificate of registration of copyright No. 78109, 04.09.2026 — software package

# **Information system for automatic detection of retinal diseases based on deep learning using optical coherence tomography images**

**8D06102 — Computer and Software Engineering**  
**Dissertation for the degree of Doctor of Philosophy (PhD)**

**Doctoral student:** *Nurmaganbet Yesmukhamedov*

**Scientific consultant:** *Cand. Sc. (Phys.-Math.), Associate Professor, IITU, Saya Sapakova*

**Foreign scientific consultant:**

*Professor, Universiti Putra Malaysia, Malaysia,  
Prof. Dr. Syed Abdul Rahman Al-Haddad*

# Limitations of the study

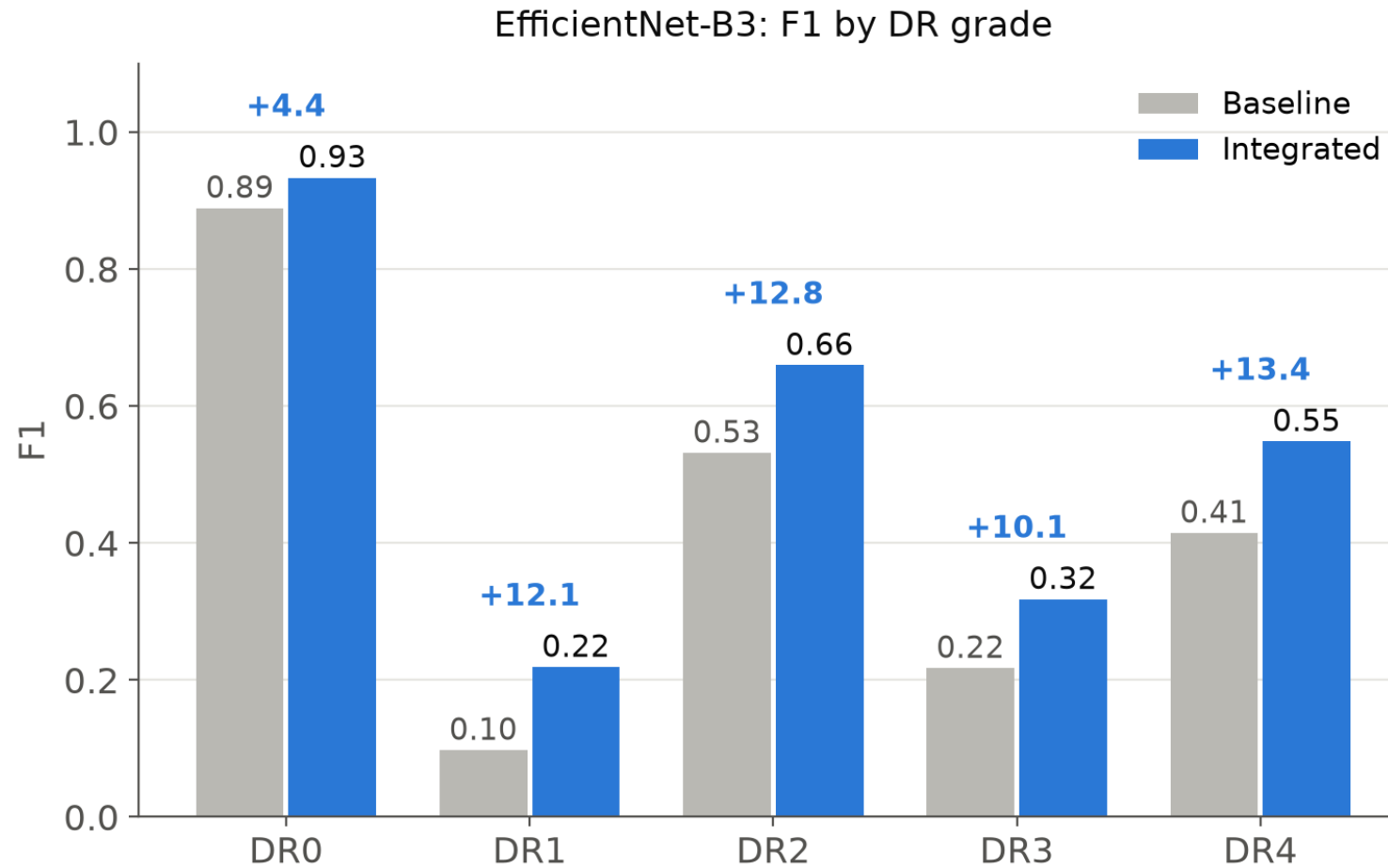


Figure 29 – EfficientNet-B3, F1 by DR grade

# Web application: dr-classification.pages.dev

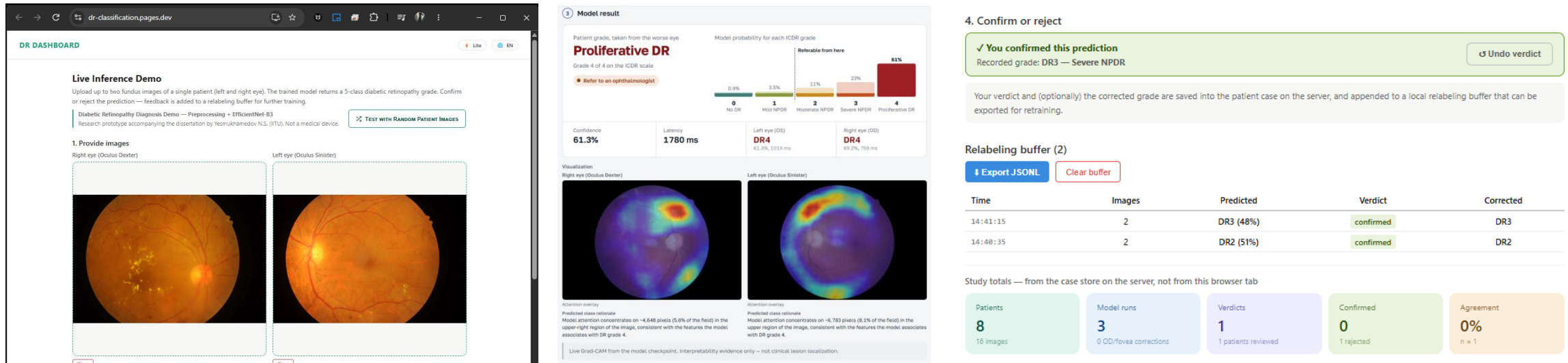


Figure 30 – Three screens of the web application: (a) uploading both eyes of one patient; (b) grade, class probabilities and Grad-CAM; (c) the doctor's confirmation and the relabelling buffer

1. **Runs in the browser:** nothing to install; Kazakh and English interface
2. **Workflow:** images of both eyes → preprocessing stages → grade, probabilities, attention map
3. **The doctor decides:** confirms or corrects the result; corrections collect in a buffer for retraining
4. **Inference** — on an RTX 3060 GPU server through a Cloudflare tunnel; a research prototype, not a medical device